

the case of power diodes and helps in finding the size of the heat-sink to be used for a given diode current and chosen type.

Electrical characteristics

In the following paragraphs, we discuss the important electrical characteristics of a semiconductor diode along with their interpretation and practical importance.

Forward voltage (V_F). It is the voltage applied across a forward-biased diode. It is not a specification in itself. It is given along with the corresponding forward current value at which it has been measured. It indicates static resistance of the diode. For example, general-purpose diode 1N3611 is specified as 1.1V at 1000 mA and it indicates that its static resistance is 1.1 ohms.

Forward current (I_F). It is the direct current flowing in a forward-biased diode.

The datasheets specify the forward voltage of the diode for one or more values of the forward current. For instance, the datasheet of diode 1N914 specifies that the maximum forward voltages of diode 1N914B are 0.72V and 1.0V at forward currents of 5.0 mA and 100 mA, respectively (Table I). This information reveals that the static resistance of the diode varies from about 140 ohms to 10 ohms as the forward current increases from 5 mA to 100 mA.

The datasheet of diode 1N4001 specifies that the forward voltage in this case is 1.1V at a forward current of 1A. The datasheets also give the forward characteristic curve of the diode showing the relationship between the forward voltage and the forward current. Fig. 5 shows the forward voltage vs forward current curve for diode 1N914B.

The information contained in the datasheet therefore allows you to know the value of the resistance offered by the diode to DC for a known value of forward current flowing through it. Said differently, it tells you the volt-

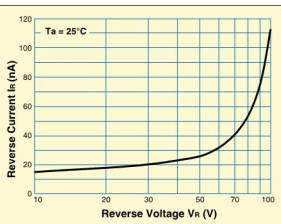


Fig. 6: Reverse-biased characteristics of 1N914B

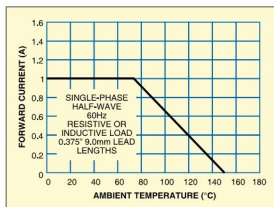


Fig. 7: Forward current derating curve of 1N4001-1N4007 series rectifiers

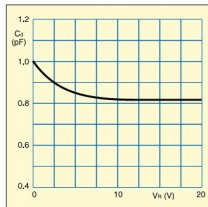


Fig. 8: Capacitance versus reverse voltage curve of 1N914

age drop across the forward-biased diode for the known value of forward current.

Reverse voltage (V_R). It is the voltage across a reverse-biased diode. It is specified along with corresponding

reverse current value at which it has been measured. It also indicates the reverse-biased resistance of the diode.

Reverse current (I_R). It is the direct current flowing through a reverse-biased diode.

The datasheets specify the reverse voltage of the diode for one or more values of the reverse current, similar to that for forward current and forward voltage.

These also specify the reverse characteristics of the diode showing the relationship between the reverse voltage and the reverse current (Fig. 6).

As is evident from the reverse characteristics of diode 1N914B shown in Fig. 6, reverse current exhibits a sharp increase as the reverse voltage exceeds 50 per cent of its peak inverse voltage of 100 volts. This implies that the reverse blocking capability of the diode de-

grades sharply as the reverse voltage approaches the peak inverse voltage rating. In addition, both forward and reverse characteristics are a function of operating temperature. Refer to the datasheet of 1N4001-1N4007 series rectifiers.

If we look at the absolute maximum ratings shown in Table II, the average rectified current is specified to be 1A up to an ambient temperature of 75°C. This parameter needs to be derated for temperatures greater than 75°C. It falls to 0.4A at 120°C (Fig. 7).

Diode capacitance (C_d , C_j). It is the inherent capacitance of the diode junction. In the reverse-bias region the transition- or the depletion-region capacitance (C_d) is of importance, whereas in the forward-bias region diffusion or storage capacitance (C_j) dominates. It plays a very significant